

Java for QA - Installation & Setup

A complete guide to prepare the environment, install, configure, validate and perform the first functional use of the tool.

Manual objective

This manual guides you through the complete installation and setup process, from preparing the environment to final validation. Follow the steps in order and use the checks throughout the guide to confirm that each configuration was completed correctly.

Official references

<https://www.oracle.com/java/technologies/downloads/>

<https://docs.oracle.com/en/java/>

How to start safely

Start from the official Java for QA site and documentation. Confirm prerequisites, supported versions, operating system and required permissions. Learn in a controlled environment before using it in a critical workflow.

First QA-oriented use

Pick one small, verifiable task in Java for QA, configure only what is necessary, run a simple example and confirm the outcome. Repeat with different data or conditions before adding abstractions.

Common issues and troubleshooting

When Java for QA fails, verify versions, dependencies, permissions, connectivity, data and environment before assuming a product defect. Preserve the original error and reduce the issue to the smallest reproducible case.

Installation overview

Download a JDK from the official source linked on this page. On Windows install it in a known directory, open a new terminal and validate with `java -version` and `javac -version`.

JAVA_HOME and PATH

If a tool requires JAVA_HOME, point it to the JDK root and add %JAVA_HOME%\bin to Path. Reopen the terminal before validating.

Forgot PATH

Open Windows Environment Variables. Add the JDK bin directory or %JAVA_HOME%\bin to Path, then open a new terminal and run `where java`, `java -version` and `javac -version`.

First validation

Create HelloWorld.java, compile with `javac HelloWorld.java` and run `java HelloWorld` to prove compiler/runtime work.

Troubleshooting

If `java -version` shows another version, use `where java` and review Path order. If `javac` is missing, verify a JDK—not only a runtime—is installed.